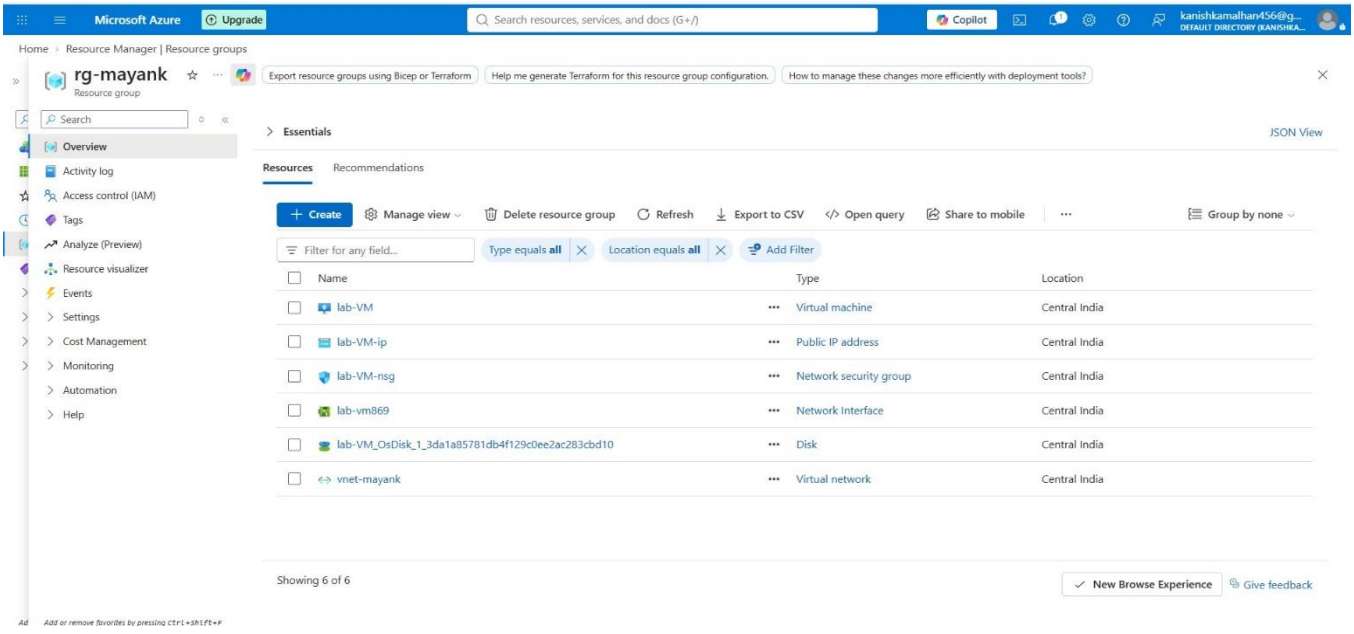
Lab-16: Azure Backup for VM Data Retention and Recovery

Objective

The objective of this lab is to configure Azure Backup for a Windows virtual machine, create a recovery point, and perform file-level recovery to verify that backed-up data can be restored successfully.

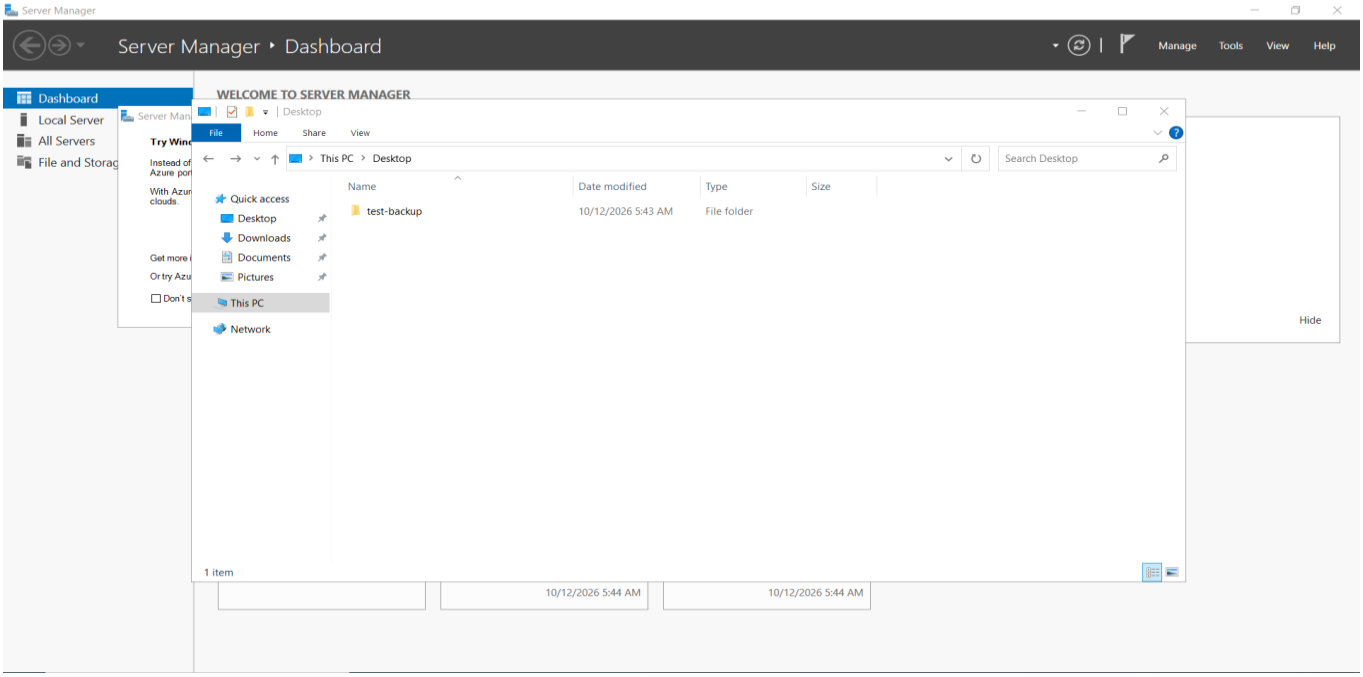
Step 1: Verify the Resource Group and VM Resources

Open the Azure Portal and navigate to the `rg-mayank` resource group. Verify that the virtual machine, public IP address, network security group, network interface, OS disk, and virtual network are available.



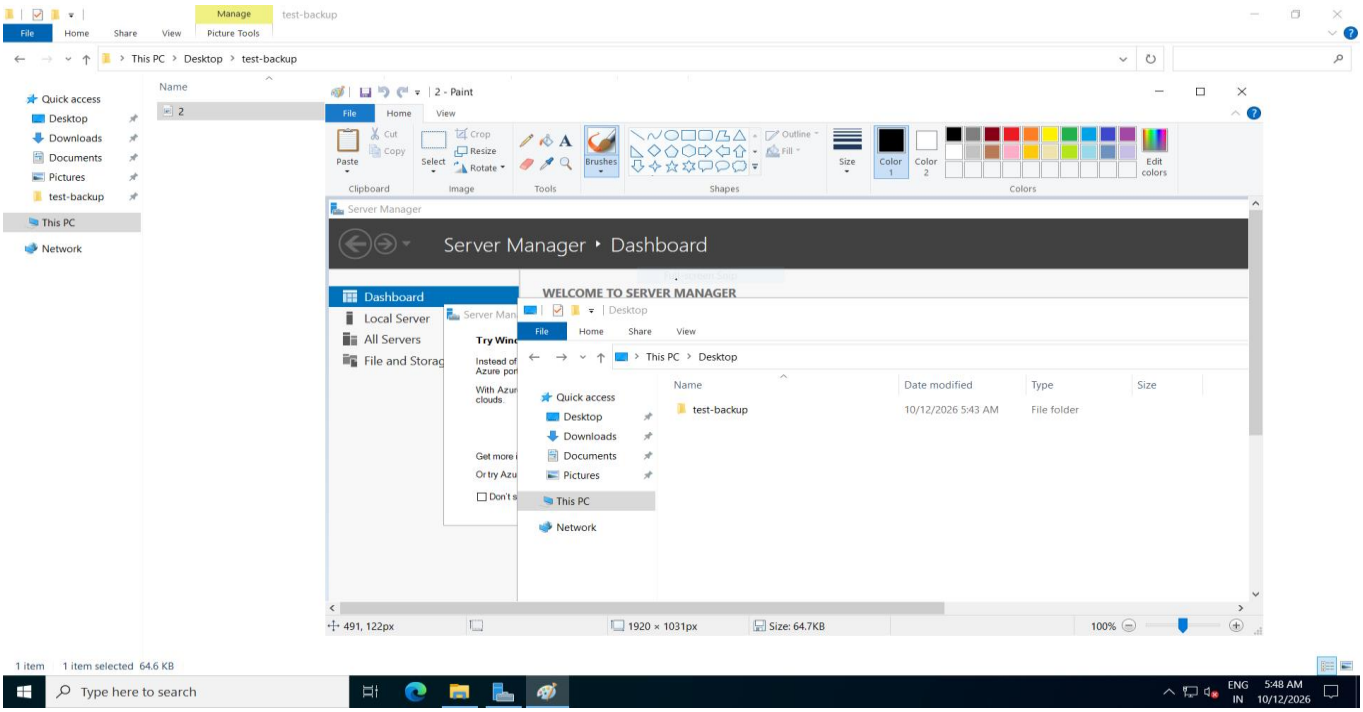
Step 2: Create a Test Folder

Connect to the Windows virtual machine using RDP. Open the Desktop and create a folder named `test-backup`. This folder will be used to verify file-level recovery.



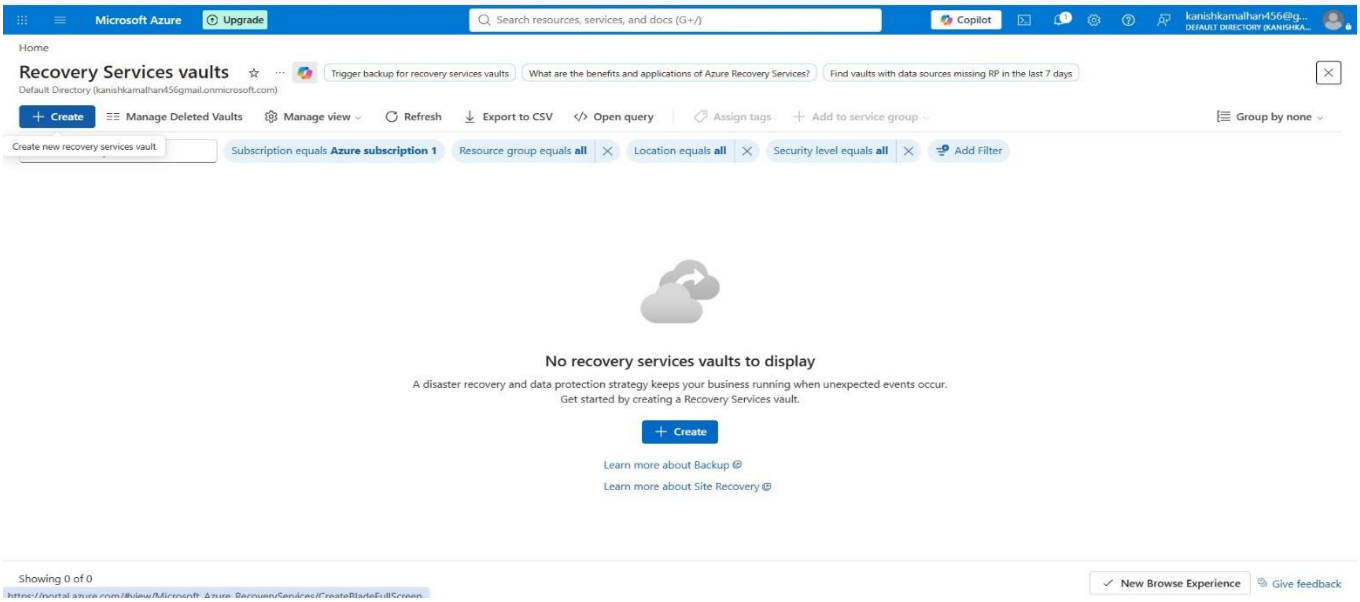
Step 3: Create Test Files

Open the `test-backup` folder and create or save sample files inside it. These files will be used to confirm that Azure Backup can recover the original data.



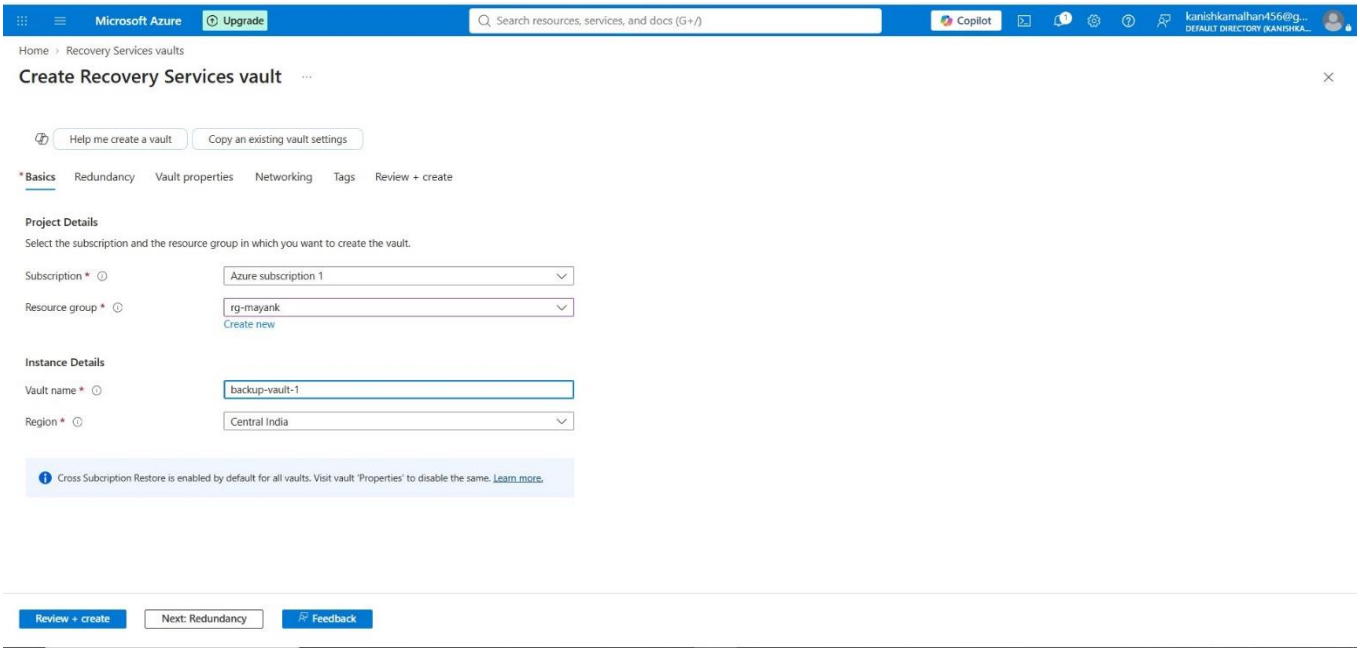
Step 4: Open Recovery Services Vaults

In the Azure Portal, search for **Recovery Services vaults**. Open the service and click **Create** to start creating a new vault.



Step 5: Configure the Recovery Services Vault

Select the subscription and resource group. Enter `backup-vault-1` as the vault name and select **Central India** as the region.



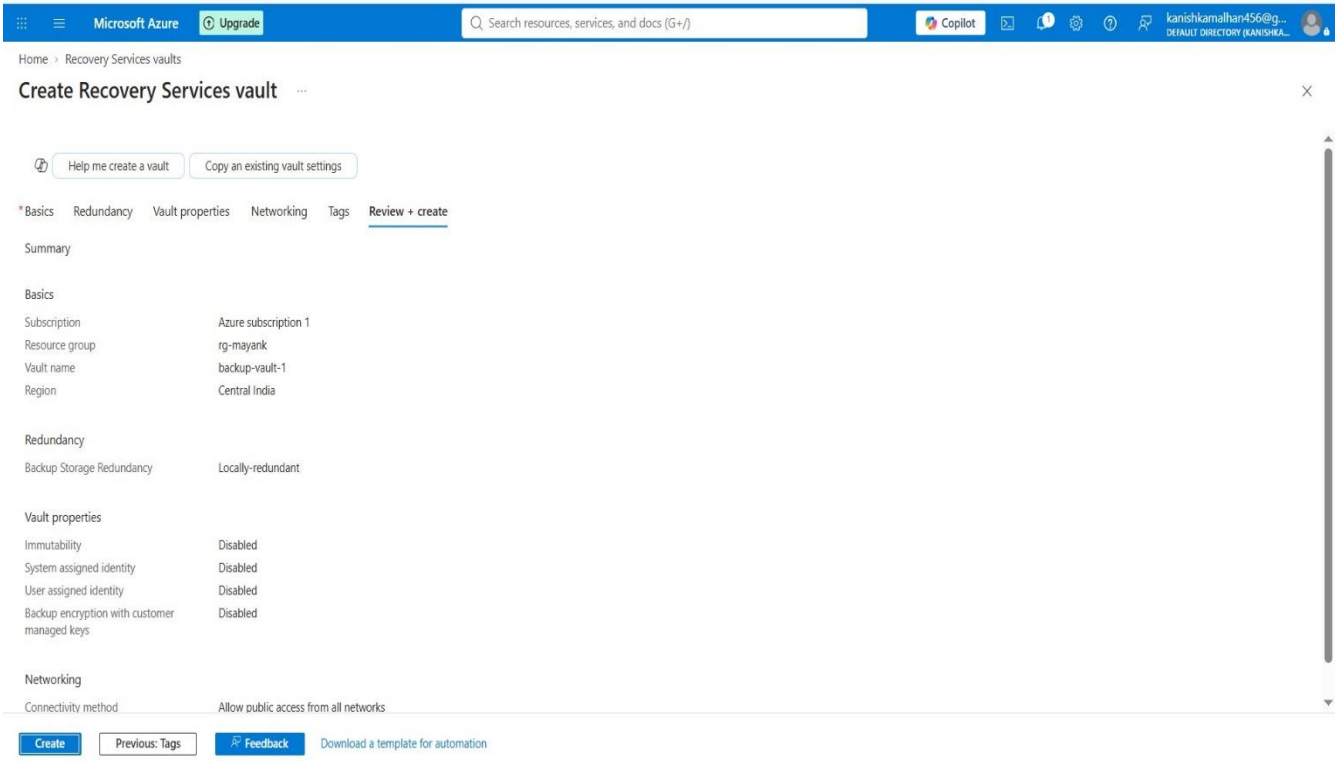
Step 6: Configure Backup Storage Redundancy

Open the **Redundancy** tab and select **Locally-redundant storage (LRS)**. This option keeps backup storage within a single Azure region and is suitable for this basic lab.



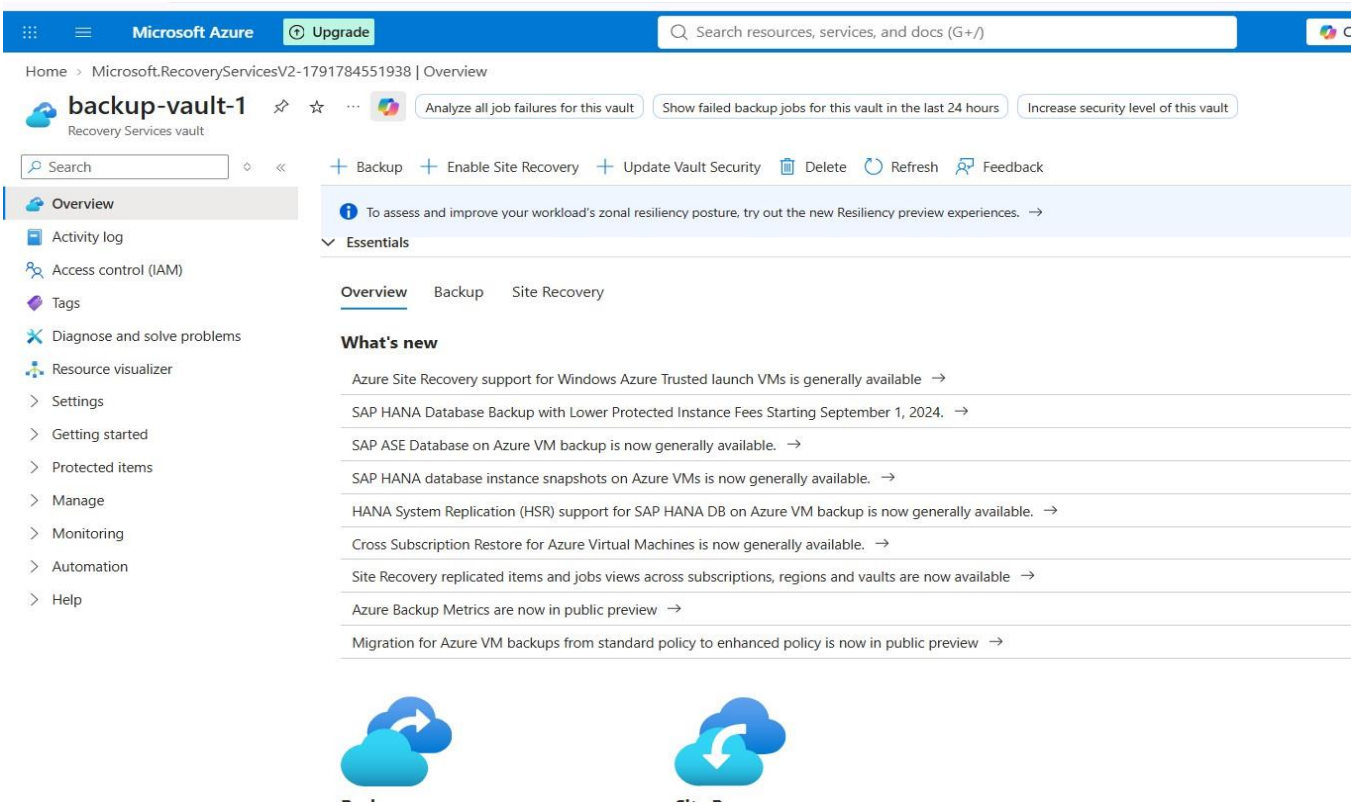
Step 7: Review and Create the Vault

Review the subscription, resource group, vault name, region, redundancy, and other configuration settings. Click **Create** to deploy the Recovery Services Vault.



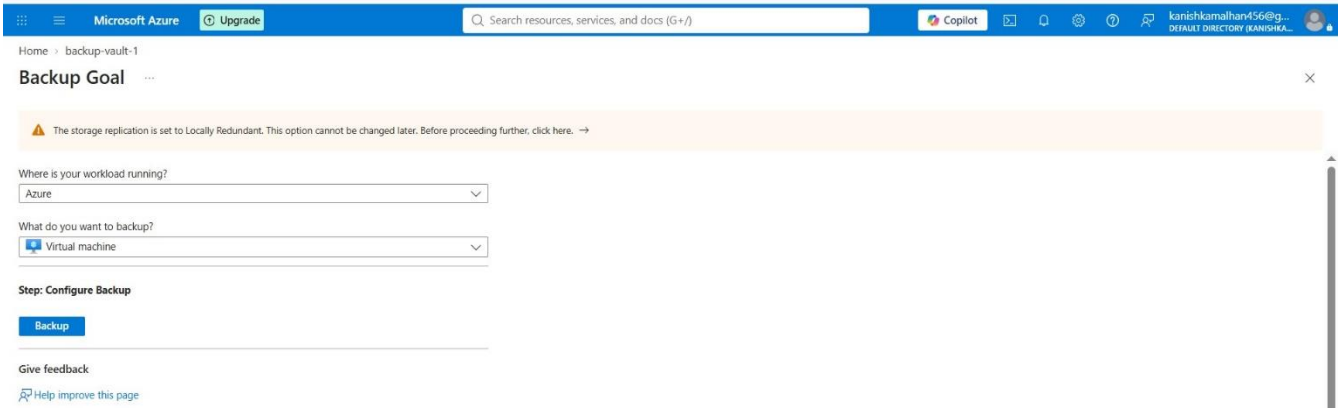
Step 8: Open the Created Vault

After deployment, open `backup-vault-1`. Verify the vault overview and locate the **Backup** option to configure protection for the virtual machine.



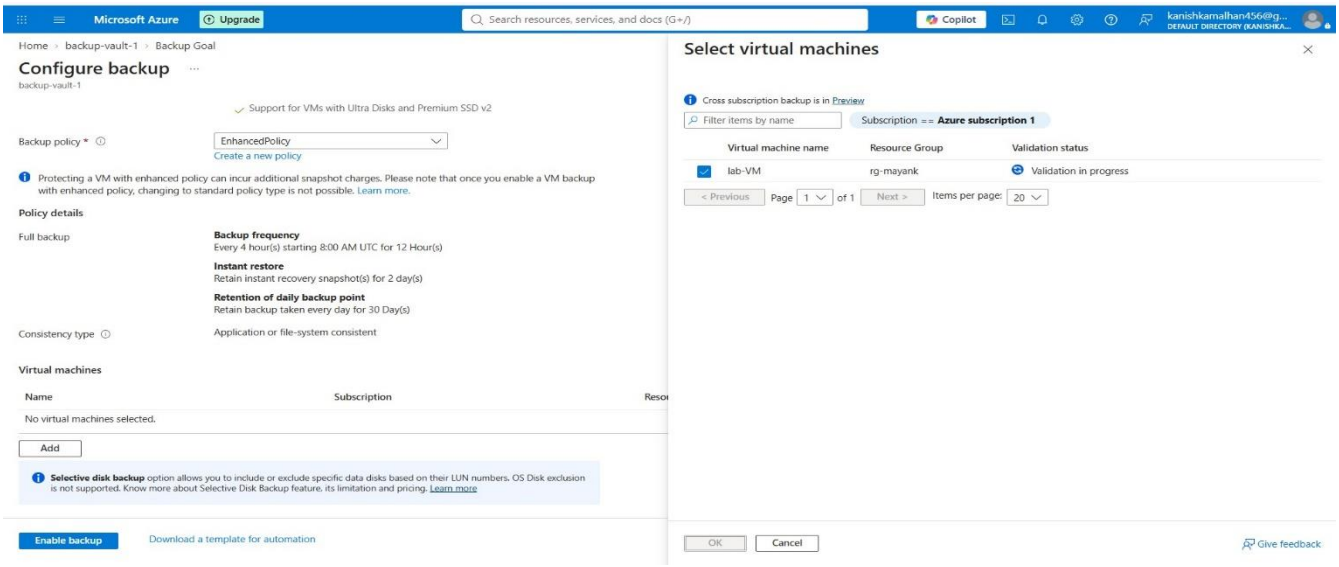
Step 9: Configure the Backup Goal

On the Backup Goal page, select **Azure** as the workload location and **Virtual machine** as the workload type. Click **Backup** to continue.



Step 10: Select the Virtual Machine

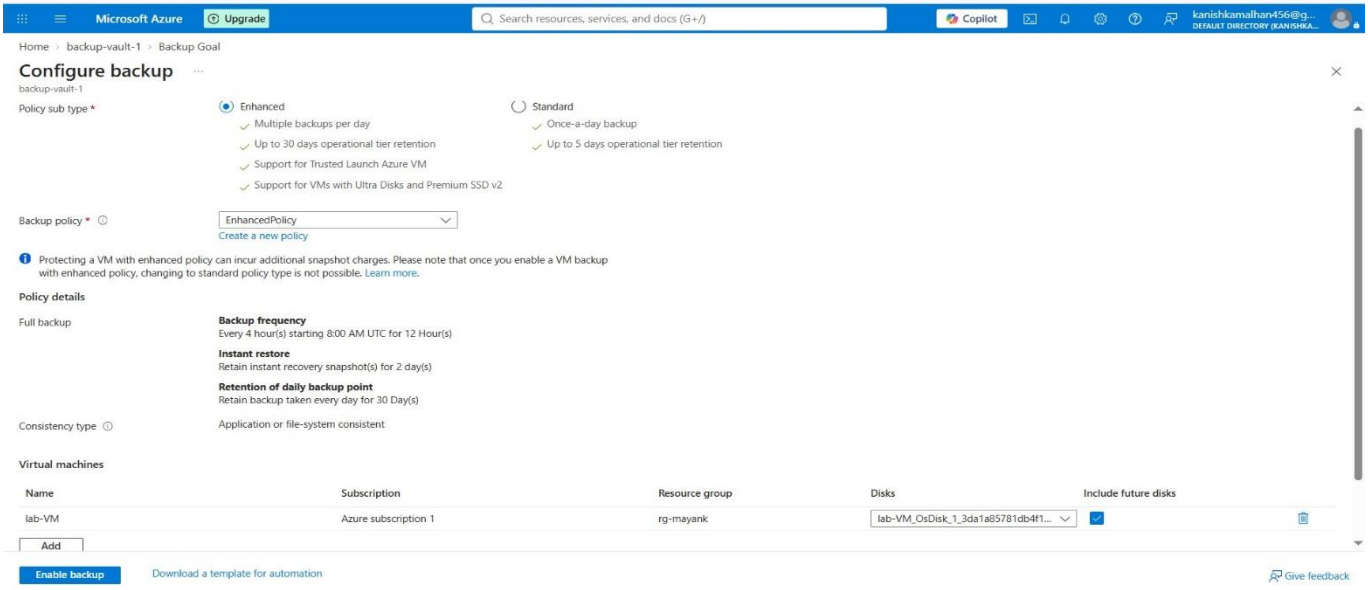
On the Configure Backup page, select the compatible backup policy and choose `lab-VM` from the available virtual machines. Wait for validation to complete before proceeding.



Step 11: Select the Enhanced Backup Policy

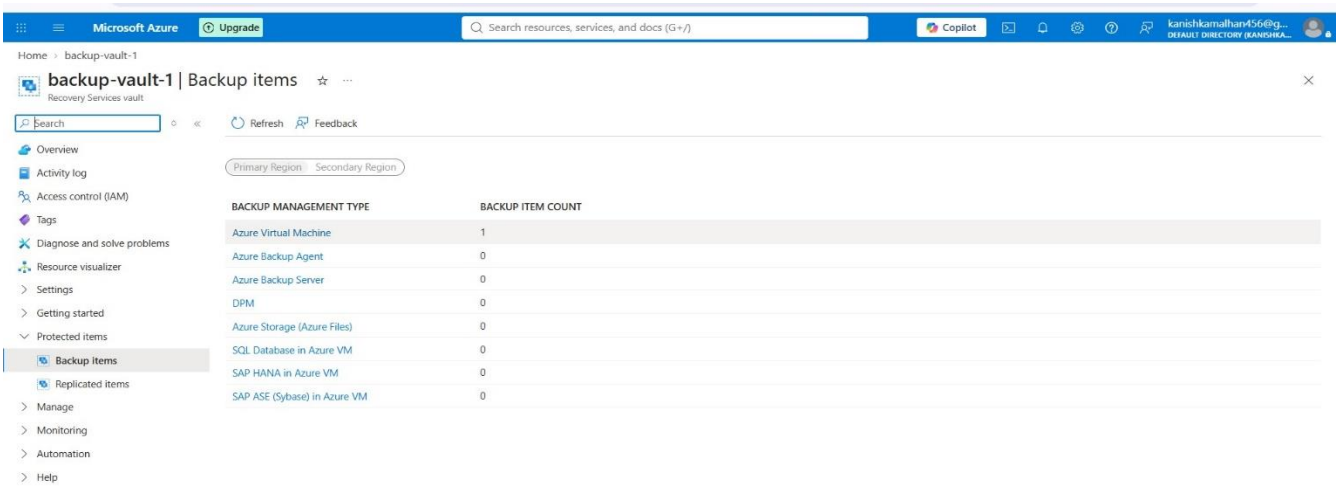
Select the **Enhanced** policy and verify the policy details, selected OS disk, and **Include future disks** setting. The Enhanced policy was used because the Standard policy was not supported for the selected Trusted Launch VM.

Click **Enable backup** to enable protection.



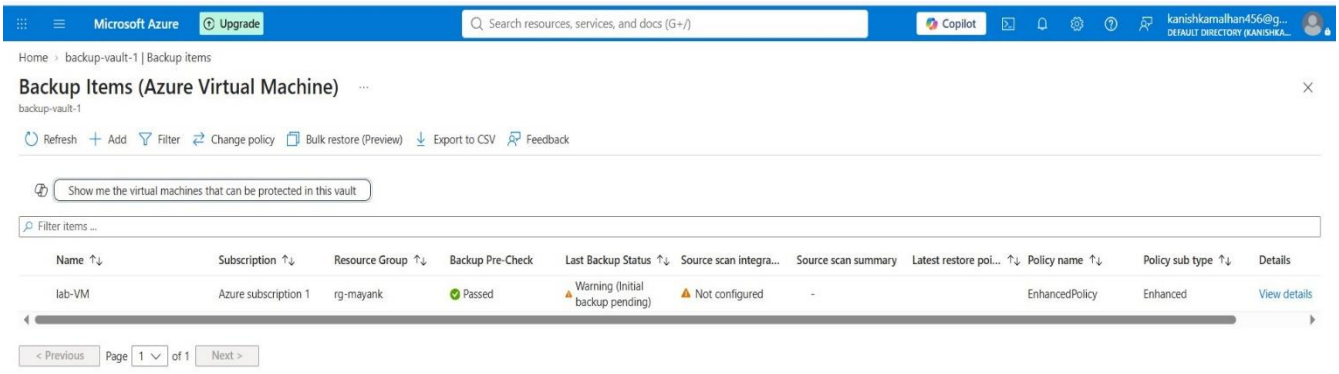
Step 12: Verify Backup Items

Open **Backup items** in the Recovery Services Vault. Select **Azure Virtual Machine** and verify that one virtual machine is listed for backup.



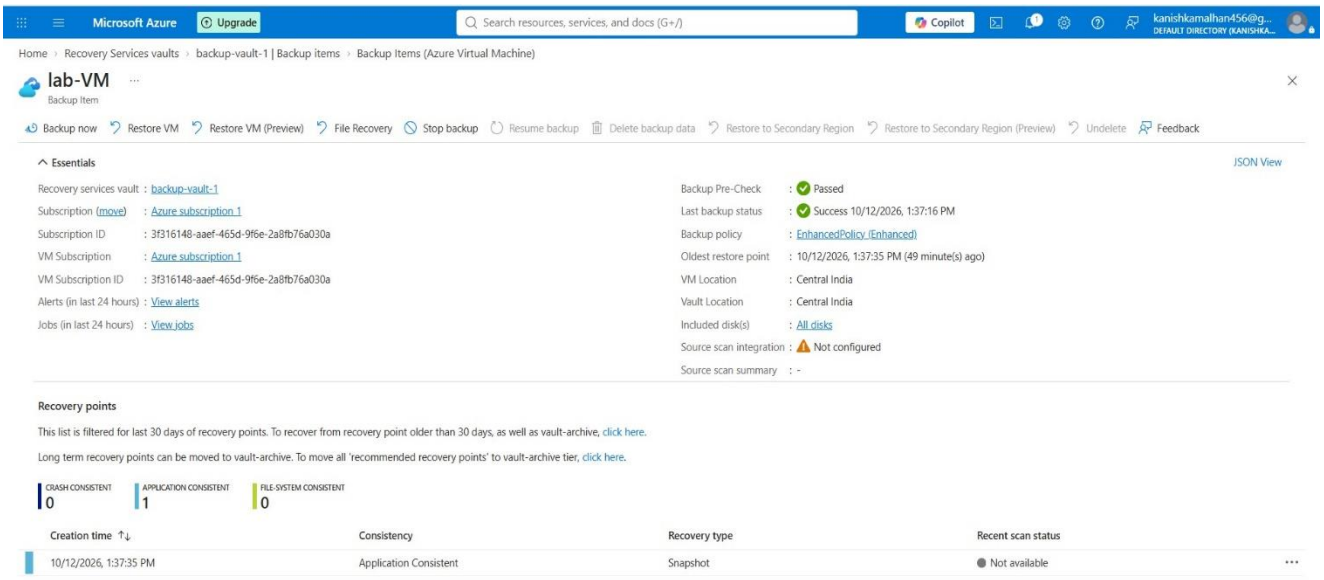
Step 13: Check the Initial Backup Status

Open the Backup Items (Azure Virtual Machine) page. Verify the VM name, backup pre-check status, selected policy, and initial backup status. If the initial backup is pending, wait for it to finish.



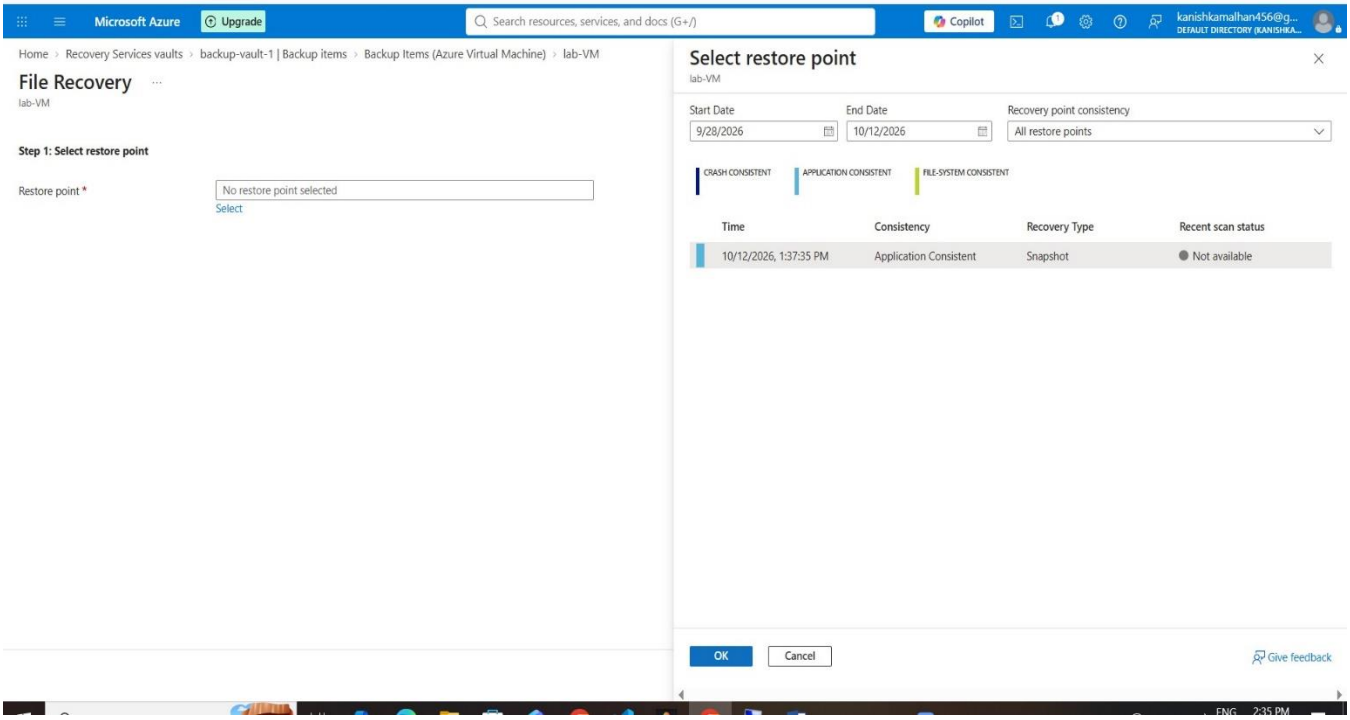
Step 14: Verify the Successful Backup and Recovery Point

Open the `lab-VM` backup item. Verify that the last backup status is **Success** and that a recovery point is available.



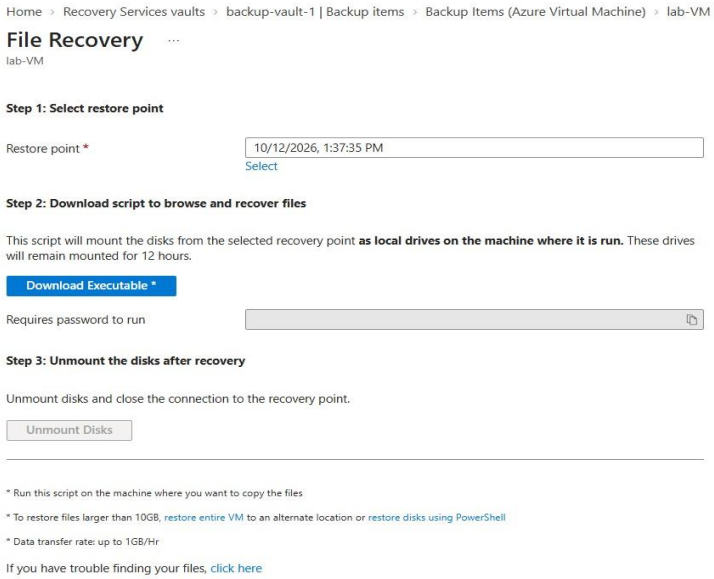
Step 15: Select a Recovery Point

Click **File Recovery** on the VM backup item. Select the required recovery point from the available list and confirm the selection.



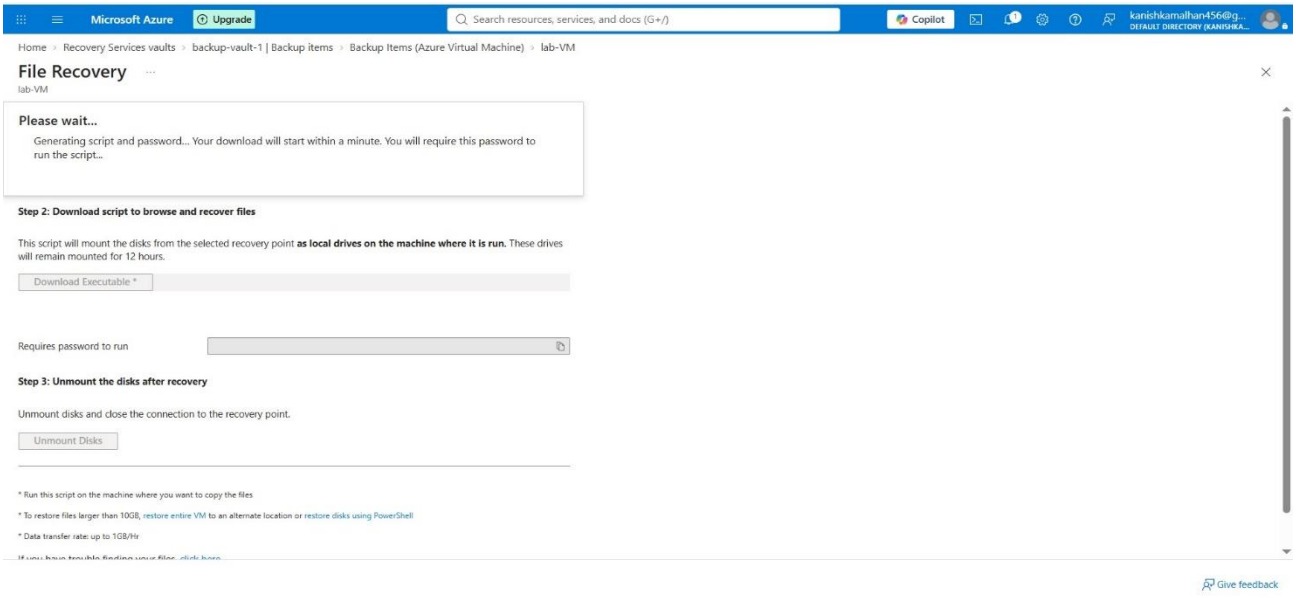
Step 16: Download the File Recovery Script

On the File Recovery page, click **Download Executable**. Wait for Azure to generate the recovery script and password required to run it.



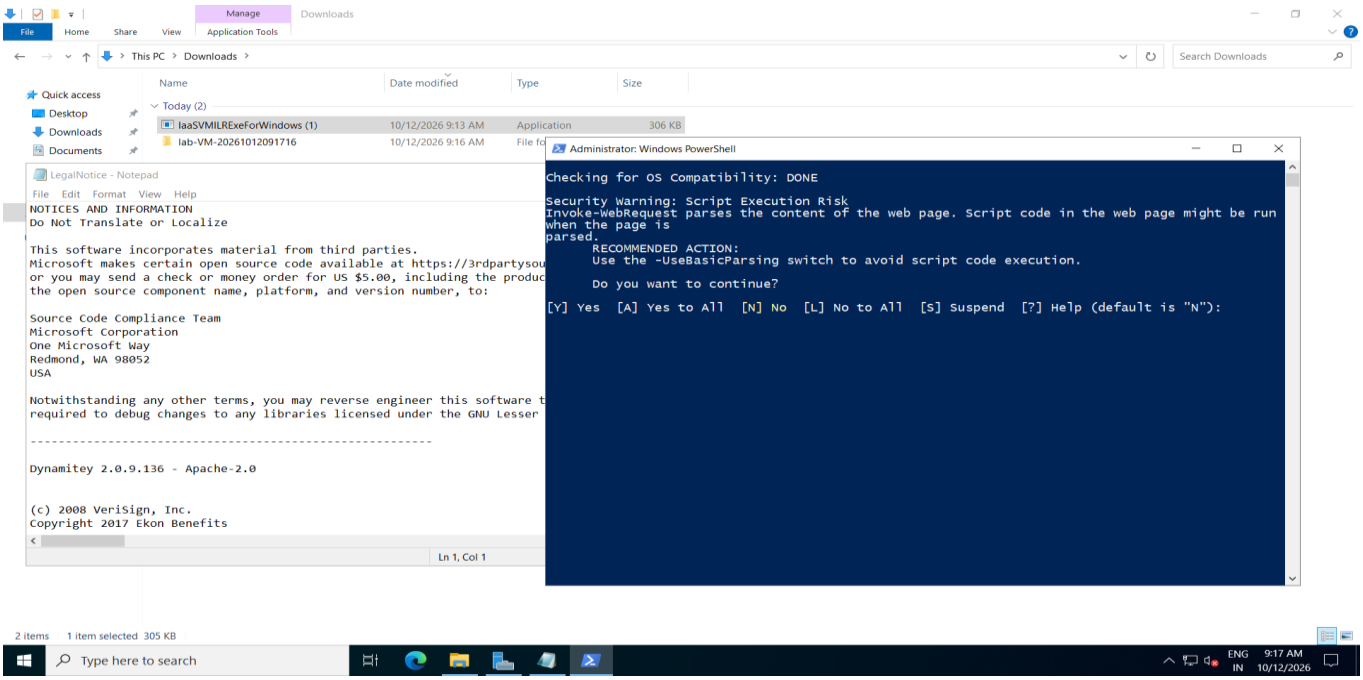
Step 17: Download and Locate the Executable

Open the Downloads folder on the Windows VM. Locate the downloaded file recovery executable and prepare to run it with administrative privileges.



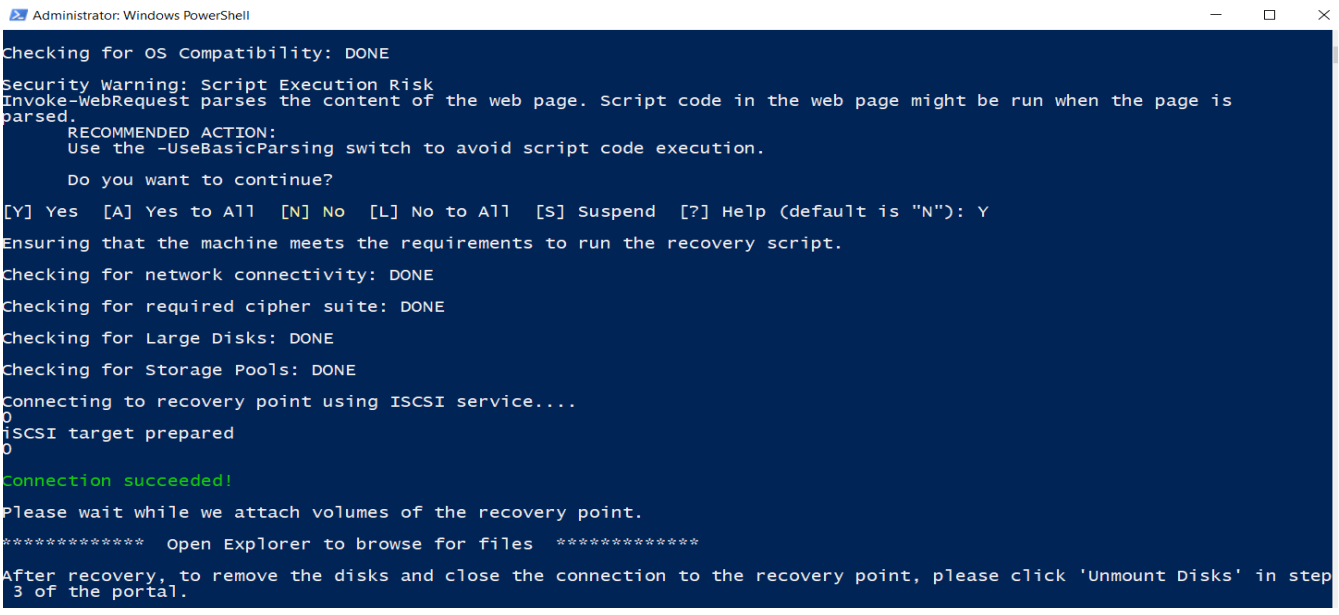
Step 18: Run the Recovery Script

Open PowerShell as Administrator and run the downloaded executable. Enter the required password and allow the script to perform its system and network checks.



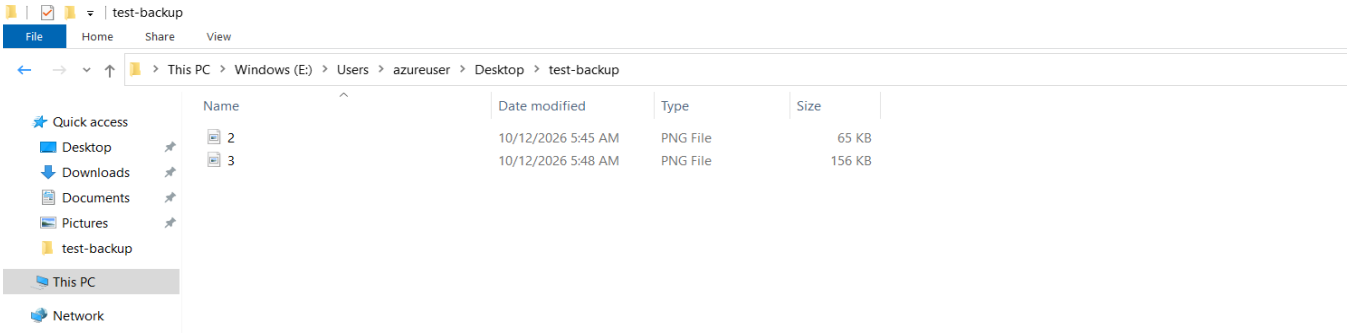
Step 19: Connect to the Recovery Point

Wait for the script to connect to the recovery point using the iSCSI service. Verify that the output reports **Connection succeeded** and that the recovery volumes have been attached.



Step 20: Verify the Recovered Files

Open File Explorer and browse the attached recovery drive. Navigate to the original location of the `test-backup` folder, locate the recovered files, and open them to verify their contents.



Result

Azure Backup was configured successfully for the Windows virtual machine. The initial backup completed, a recovery point was created, and file-level recovery was performed successfully. The recovered files were located and verified.